

Smooth Kernels Suffice for Two-Layer Universal Operator Approximation

Candidate full solution for arXiv:1910.01545

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Status: full solution, likely valid, pending human review. The theorem below affirmatively answers the existence part of the first future-work question in Section 6 of Guss–Salakhutdinov, and removes the suggested extra regularity assumption on the target operator.

1 Source problem and scope

The source is William H. Guss and Ruslan Salakhutdinov, *On Universal Approximation by Neural Networks with Uniform Guarantees on Approximation of Infinite Dimensional Maps*, arXiv:1910.01545v1 (2019) [1]. On page 12, Section 6, the authors ask:

“Given some regularity conditions on a nonlinear operator, are the weight functions of the operator layer based neural networks that approximate it smooth or regular?”

Figure 1 reproduces the source context. We answer the existence question in the affirmative. In fact, continuity of the nonlinear operator and compactness of the input family already suffice; no additional regularity of the operator is needed.

Our results suggest directions for future work. First, we show that infinite dimensional neural networks composed of operator layers are universal approximators of nonlinear operators, and in addition we provide sufficient conditions for some finite dimensional neural network to approximate an infinite dimensional one. A natural next question is: given some regularity conditions on a nonlinear operator, are the weight functions of the operator layer based neural networks that approximate it smooth or regular? An answer in the affirmative yields a new method for parameterizing finite neural networks by learning smooth estimators of these weight functions with upper bounds on how many samples are needed to achieve universal approximation. Second, the uniform universality relationship between finite dimensional and infinite dimensional neural networks opens the possibility of approaching the problem of non convex optimization of neural networks using techniques from the calculus of variations on their infinite dimensional analogues to aid in convergence results. Finally, there is a question of when our upper bound on the minimum number of input and output units required for uniformity can be strengthened. In particular, we leave the characterization of domains and function spaces for which the bound becomes sub-exponential to future work.

Figure 1: The first future-work question in Section 6, page 12 of arXiv:1910.01545v1.

The conclusion is qualitative and existential. It does not provide bounds on derivatives, sample complexity, or the outcome of a training algorithm.

2 Set-up and exact statement

Let $K \subset \mathbb{R}^d$ and $K' \subset \mathbb{R}^{d'}$ be the compact domains of the source paper, equipped with Lebesgue measure. Let $C(K)$ and $C(K')$ carry the supremum norm. We say that a function on a compact Euclidean set is *ambient smooth* if it is the restriction of a C^∞ function on the surrounding Euclidean space. Our construction gives compactly supported ambient extensions.

For a latent domain $I = [0, 1]$, smooth kernels and biases define the two-layer network

$$\mathcal{N}(f)(y) = \int_I W_2(v, y) g\left(\int_K f(u) W_1(u, v) du + B_1(v)\right) dv + B_2(y). \quad (1)$$

This is precisely a composition of two affine operator layers with a pointwise activation between them.

Theorem 1 (smooth two-operator-layer universality). *Let $K \subset \mathbb{R}^d$ and $K' \subset \mathbb{R}^{d'}$ be compact domains, let $E \subset C(K)$ be compact, and let $F : C(K) \rightarrow C(K')$ be continuous. If $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and nonpolynomial, then for every $\varepsilon > 0$ there exist*

$$W_1 \in C_c^\infty(\mathbb{R}^d \times \mathbb{R}), \quad W_2 \in C_c^\infty(\mathbb{R} \times \mathbb{R}^{d'}),$$

and $B_1 \in C_c^\infty(\mathbb{R})$, $B_2 \in C_c^\infty(\mathbb{R}^{d'})$ such that the network (1), using their restrictions to $K \times I$, $I \times K'$, I , and K' , satisfies

$$\sup_{f \in E} \|\mathcal{N}(f) - F(f)\|_\infty < \varepsilon.$$

Consequently, two-layer operator networks whose two weight kernels and two biases are jointly ambient C^∞ are dense, on every compact input family, in $C(C(K), C(K'))$.

3 Proof intuition

The key is that the source paper has already reduced the infinite-dimensional problem to finitely many hidden features. Its Theorem 2 gives

$$N_0(f)(y) = \sum_{i=1}^N a_i(y) g\left(\int_K f(u) c_i(u) du + b_i\right) + a_0(y),$$

where $c_i \in L^1(K)$ and $a_i, a_0 \in C(K')$. On the compact input set E , the input norms are uniformly bounded. Thus small L^1 changes in c_i make all preactivations uniformly close. The activation is uniformly continuous on the single compact interval containing both the original and perturbed preactivations. Smooth density in L^1 and Stone–Weierstrass therefore smooth all coefficients without losing uniform accuracy.

The resulting expression is still a finite hidden sum. To express it *exactly* with two integral operator layers, assign one disjoint smooth bump to each hidden unit in the latent interval. A companion cutoff is one on that bump’s support, so the latent integral reads each hidden feature without cross-talk. The integral of each normalized bump is one, hence the integral is exactly the finite sum.

4 Proof

Proof of Theorem 1. Apply Theorem 2 of the source paper [1] with tolerance $\varepsilon/2$. There are $N \geq 1$, functions $c_i \in L^1(K)$, $a_i \in C(K')$, constants $b_i \in \mathbb{R}$, and $a_0 \in C(K')$ such that

$$\sup_{f \in E} \|N_0(f) - F(f)\|_\infty < \frac{\varepsilon}{2}, \quad (2)$$

where

$$N_0(f)(y) = \sum_{i=1}^N a_i(y)g(z_i(f)) + a_0(y), \quad z_i(f) = \int_K f(u)c_i(u) du + b_i.$$

Step 1: smooth the finite-hidden network. Compactness of E gives

$$M := \max_{f \in E} \|f\|_\infty < \infty.$$

Put

$$A := \sum_{i=1}^N \|a_i\|_\infty, \quad R := 1 + \max_{1 \leq i \leq N} (M \|c_i\|_{L^1} + |b_i|), \quad G := \max_{|t| \leq R} |g(t)|.$$

By uniform continuity of g on $[-R, R]$, choose $0 < \eta < 1$ so that

$$|g(s) - g(t)| < \frac{\varepsilon}{8(1+A)} \quad \text{if } s, t \in [-R, R] \text{ and } |s - t| < \eta. \quad (3)$$

Extend each c_i by zero outside K . Density of $C_c^\infty(\mathbb{R}^d)$ in $L^1(\mathbb{R}^d)$ gives $p_i \in C_c^\infty(\mathbb{R}^d)$ with

$$\|p_i - c_i\|_{L^1(K)} < \frac{\eta}{M+1}. \quad (4)$$

For every $f \in E$,

$$\left| \int_K f(u)(p_i - c_i)(u) du \right| \leq M \|p_i - c_i\|_{L^1(K)} < \eta.$$

The original preactivations have absolute value at most $R - 1$; hence the new ones

$$z'_i(f) := \int_K f(u)p_i(u) du + b_i$$

belong to $[-R, R]$. Therefore (3) applies uniformly in f .

By the real Stone–Weierstrass theorem, polynomial restrictions are uniformly dense in $C(K')$. Multiplying the approximating polynomials by a smooth cutoff equal to one on a neighborhood of K' shows that restrictions of $C_c^\infty(\mathbb{R}^{d'})$ are uniformly dense in $C(K')$. Choose $q_i, q_0 \in C_c^\infty(\mathbb{R}^{d'})$ such that

$$\|q_i - a_i\|_{C(K')} < \frac{\varepsilon}{8N(1+G)}, \quad \|q_0 - a_0\|_{C(K')} < \frac{\varepsilon}{8}. \quad (5)$$

Define the smoothed finite-hidden network

$$N_s(f)(y) := \sum_{i=1}^N q_i(y)g(z'_i(f)) + q_0(y).$$

Adding and subtracting $a_i(y)g(z'_i(f))$, and using (3)–(5), yields

$$\begin{aligned}\|N_s(f) - N_0(f)\|_\infty &\leq \sum_{i=1}^N \|q_i - a_i\|_\infty G + \sum_{i=1}^N \|a_i\|_\infty \frac{\varepsilon}{8(1+A)} + \|q_0 - a_0\|_\infty \\ &< \frac{\varepsilon}{8} + \frac{\varepsilon}{8} + \frac{\varepsilon}{8} = \frac{3\varepsilon}{8}.\end{aligned}$$

Together with (2), this gives

$$\sup_{f \in E} \|N_s(f) - F(f)\|_\infty < \frac{7\varepsilon}{8} < \varepsilon. \quad (6)$$

Step 2: encode the finite sum exactly by smooth operator layers. Choose pairwise disjoint open intervals $J_1, \dots, J_N$ compactly contained in $(0, 1)$. For each i , choose a nonnegative $\zeta_i \in C_c^\infty(J_i)$ with $\int_0^1 \zeta_i(v) dv = 1$. Choose $\chi_i \in C_c^\infty(J_i)$ that equals one on $\text{supp } \zeta_i$. Set

$$\begin{aligned}W_1(u, v) &:= \sum_{i=1}^N \chi_i(v) p_i(u), & B_1(v) &:= \sum_{i=1}^N \chi_i(v) b_i, \\ W_2(v, y) &:= \sum_{i=1}^N \zeta_i(v) q_i(y), & B_2(y) &:= q_0(y).\end{aligned}$$

These functions have the asserted compactly supported smooth ambient extensions. If $v \in \text{supp } \zeta_i$, disjointness gives $\chi_i(v) = 1$ and $\chi_j(v) = 0$ for $j \neq i$. Consequently,

$$\begin{aligned}&\int_0^1 W_2(v, y) g\left(\int_K f(u) W_1(u, v) du + B_1(v)\right) dv + B_2(y) \\ &= \sum_{i=1}^N q_i(y) \int_0^1 \zeta_i(v) g\left(\int_K f(u) p_i(u) du + b_i\right) dv + q_0(y) \\ &= \sum_{i=1}^N q_i(y) g(z'_i(f)) + q_0(y) = N_s(f)(y).\end{aligned}$$

The operator-layer network is therefore exactly N_s , and (6) proves the claim. $\square$

5 Verification notes

1. **Topology check.** The only use of compactness of E is the uniform input bound M . Continuity of F and nonpolynomiality of g enter through the source paper’s Theorem 2.
2. **Norm check.** L^1 approximation of c_i is the correct norm because inputs are bounded in L^∞ ; Hölder’s inequality gives the uniform preactivation bound.
3. **Activation check.** No global Lipschitz assumption on g is used. Both old and new preactivations lie in $[-R, R]$, where continuity implies uniform continuity and boundedness.
4. **Joint smoothness check.** Each kernel is a finite sum of products of ambient smooth functions in separate variables, hence is jointly C^∞ . All supports are compact.
5. **Architecture check.** The disjoint supports eliminate all cross-terms, and normalization $\int \zeta_i = 1$ makes the latent integral exactly the finite hidden sum.
6. **Mechanical check.** The script `code/verify_error_budget.py` verifies that the allocated errors sum to $7\varepsilon/8$ and checks the diagonal overlap algebra of the bump encoding using exact rational arithmetic.

6 Limitations and novelty check

The theorem answers whether smooth weights *can be chosen* in uniformly approximating operator-layer networks. It does not show that every representation is smooth, that training discovers such weights, or that an exact representation of F exists. It is qualitative: no derivative bounds, rates, network-size estimates, or sample bounds are claimed. The other two future directions in the source conclusion (variational optimization and sub-exponential sampling bounds) are untouched.

A bounded novelty search was performed on August 11, 2026. The run’s cheap registry, solution, attempt, and proof-gap indexes were searched for the arXiv identifier and the core phrases. The official arXiv API was queried for “operator layer” with “smooth weight”, “neural operator” with “smooth kernel” and “universal approximation”, and the exact source title. Bounded arXiv web searches used close variants. They found later neural-operator universality results, notably Kovachki et al. [2] and a KNO theorem with continuous positive-definite kernels [3], but no theorem located in those bounds states the present ambient-smooth, two-layer, $C(K) \rightarrow C(K')$, uniform-on-compact-family conclusion. This is a bounded search, not an exhaustive literature review; novelty confidence remains moderate.

7 Human-review recommendation

The argument is short, uses only standard density results, and preserves the source architecture exactly. Review should focus on (i) matching the source’s “compact domains” convention for Lebesgue integration, (ii) confirming that the future-work question is intended existentially rather than as a request for quantitative regularity inherited from F , and (iii) checking for an unindexed later statement of the same smoothing observation. Mathematically, the packet is recommended for full-solution review.

References

- [1] W. H. Guss and R. Salakhutdinov, *On Universal Approximation by Neural Networks with Uniform Guarantees on Approximation of Infinite Dimensional Maps*, arXiv:1910.01545v1, 2019.
- [2] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar, *Neural Operator: Learning Maps Between Function Spaces*, arXiv:2108.08481.
- [3] M. Lowery, J. Turnage, Z. Morrow, J. D. Jakeman, A. Narayan, S. Zhe, and V. Shankar, *Kernel Neural Operators (KNOs) for Scalable, Memory-efficient, Geometrically-flexible Operator Learning*, arXiv:2407.00809.
- [4] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987. Standard references for L^1 approximation and Stone–Weierstrass.